



ANALYSTLAB AFRICA
Experience Lab Internship Programme
Official Weekly Assignment
Week 1: FinTrust Digital Bank — Understand & Plan

Document Control

Week	Week 1
Project	FinTrust Financial Intelligence & Digital Banking Support Solution
Version	1.0
Issue Date	
Submission Deadline	End of Week 1 (Sunday, 11:59 PM WAT)

1. Week 1 Overview

Welcome to **Week 1 of the FinTrust Digital Bank Experience Lab Project.**

FinTrust Digital Bank is a fictional digital banking organisation seeking to use **data, machine learning and generative AI** to improve its understanding of customer and transaction behaviour, support risk-related decision-making, improve operational processes and strengthen digital customer support.

Over the next four weeks, you will work within your assigned track to contribute to the development of an integrated FinTrust solution.

Week 1 is about building the foundation.

Before developing dashboards, machine-learning models, APIs or AI assistants, you must first understand:

- the business problem;
- the available resources;
- your track's role;
- the requirements of your solution;
- the questions your work should answer;
- how success will be measured; and
- what you intend to build during Weeks 2–4.

Important: There is **no formal cross-track collaboration requirement in Week 1**. You are expected to complete your Week 1 work independently within your assigned track.

Formal cross-track collaboration will begin in **Week 2**, when interns have produced tangible outputs that can be exchanged and used by other tracks.

2. Project Resources

You must use the approved FinTrust resources provided by AnalystLab Africa:

Data Resources

- [FinTrust Customer Data.csv](#)
- [FinTrust Transaction Data.csv](#)
- [FinTrust Data Dictionary.xlsx](#)

GenAI Resource

- [FinTrust Financial Knowledge Base.docx](#)

Project Resources

- [FinTrust Project Brief](#)
- [FinTrust 4-Week Master Roadmap](#)

Data Disclaimer

All FinTrust customers, transactions and financial information are **synthetic and created for educational purposes**.

3. General Week 1 Requirements

Regardless of your track, you must complete the following:

A. Understand the Business Problem

Explain:

1. What problem is FinTrust trying to solve?
2. Why is the problem important?
3. How can your track contribute to solving it?
4. What type of output could your track provide?

B. Review the Relevant Resources

Identify the resources relevant to your track and explain:

- what each resource contains;
- how you expect to use it;
- any limitations you have identified.

C. Define Your Track Objectives

State **3–5 specific objectives** for your work on the FinTrust project.

D. Identify Success Criteria

Explain how you will determine whether your eventual solution is successful.

E. Develop an Initial Plan

Create a clear plan showing what you expect to accomplish during:

- Week 2
- Week 3
- Week 4

Your plan may change later as you learn more. Week 1 is about establishing a strong starting point.

4. DATA ANALYTICS TRACK

Task: FinTrust Business & Data Intelligence Assessment

Your role is to determine how FinTrust's customer and transaction data can be transformed into useful business intelligence.

Part A — Business Understanding

Answer:

1. What business questions should FinTrust's management be able to answer?
2. What decisions could data analysis support?
3. What stakeholders could benefit from the analytical outputs?

Part B — Data Understanding

Review both datasets.

Document:

- number of records;
- number of columns;
- field names;
- data types;
- categorical variables;
- numerical variables;

- date/time variables;
- relationship between the datasets;
- missing values;
- obvious data-quality issues.

Do not perform extensive data cleaning yet.

Part C — Analytical Questions

Develop **5–8 meaningful analytical questions**.

Your questions should cover areas such as:

- customer behaviour;
- transaction activity;
- transaction value;
- transaction channels;
- transaction status;
- risk-related patterns.

Avoid questions that cannot realistically be answered using the supplied data.

Part D — KPI Planning

Propose **5–8 KPIs**.

Use this structure:

KPI Definition Why It Matters Required Data

Part E — Dashboard Planning

Create a **dashboard wireframe** showing:

- proposed dashboard sections;
- key KPI cards;
- charts/visuals;
- filters;
- major insights you expect the dashboard to communicate.

A fully developed Power BI dashboard is **not required in Week 1**.

Recommended Tools

- Excel
- SQL
- Python/Pandas
- Power BI

Use tools appropriately rather than forcing every tool into the assignment.

DA Deliverables

1. Business Understanding
2. Data Understanding/Profiling Report
3. Analytical Questions
4. KPI Definition Table
5. Dashboard Wireframe
6. Initial Analysis Plan

5. DATA SCIENCE TRACK

Task: FinTrust Predictive Intelligence Problem Definition

Your role is to determine how machine learning could support FinTrust's risk-related decision-making.

Part A — Predictive Problem

Define a potential machine-learning problem for FinTrust.

Explain:

- what you want to predict;
- why prediction could be useful;
- who could use the prediction;
- what a successful model would need to achieve.

Part B — Target Assessment

Investigate:

Risk_Review_Flag

Determine:

- what the field represents;
- its possible values;
- whether it can serve as a modelling target;
- whether the target appears balanced or imbalanced;
- potential limitations of using it.

Remember that this is a **synthetic educational label**, not a real fraud determination.

Part C — Candidate Features

Identify at least **8 potential features**.

Feature Why It May Matter Data Type Potential Concern

Part D — Hypotheses

Develop at least **3 hypotheses** about factors that may be associated with the risk-review label.

Do not present the hypotheses as conclusions.

Part E — Modelling Plan

Create an initial plan covering:

- data preparation;
- exploratory analysis;
- feature engineering;
- baseline model;
- candidate algorithms;
- evaluation metrics;
- validation;
- error analysis;
- model limitations.

Recommended Tools

- Python
- Pandas
- NumPy
- Scikit-learn
- Jupyter Notebook/Google Colab
- Git/GitHub

DS Deliverables

1. Predictive Problem Statement
2. Target Assessment
3. Feature Candidate Table
4. Three or more Hypotheses
5. Initial Modelling Plan
6. Data-Quality Observations

6. MACHINE LEARNING ENGINEERING TRACK

Task: FinTrust ML Solution Architecture & Technical Planning

Your role is to plan how a future machine-learning solution can move from data and model development into a reproducible technical workflow.

Part A — Understand the ML Requirement

Review the FinTrust Project Brief and datasets.

Explain:

- what the ML component is expected to accomplish;
- what inputs it may require;
- what output it should produce.

Part B — ML Workflow

Design a high-level workflow:

Data → Validation → Preprocessing → Feature Preparation → Model → Prediction → Output → Testing

Explain the purpose of each stage.

Part C — Technical Requirements

Identify:

- input requirements;
- output requirements;
- preprocessing requirements;
- dependencies;
- testing requirements;
- version-control requirements;
- potential API/service requirements.

Part D — Repository Structure

Design a professional repository structure.

Explain the purpose of each folder you include.

Part E — Technical Risks

Identify at least **5 potential technical risks**.

For example:

- inconsistent data;
- missing values;
- model input mismatch;
- dependency problems;
- integration failure;
- reproducibility problems.

Recommended Tools

- Python
- Git
- GitHub
- Jupyter/VS Code
- API concepts
- Testing concepts

MLE Deliverables

1. ML Architecture Diagram
2. Technical Requirements Document
3. Repository Structure
4. Technical Risk List
5. Initial Testing Strategy
6. Implementation Plan

7. GENERATIVE AI TRACK

Task: FinTrust AI Assistant Scope & Knowledge Assessment

Your role is to determine how Generative AI can provide useful customer support while remaining grounded in approved FinTrust information.

Part A — Knowledge Base Review

Read:

FinTrust_Financial_Knowledge_Base.docx

Identify:

- supported topics;
- available information;
- unsupported information;
- escalation situations;
- sensitive information.

Part B — Assistant Use Cases

Identify at least **5 appropriate use cases**.

Then identify at least **5 things the assistant should NOT do**.

Part C — Initial Prompt Design

Develop prompts for scenarios such as:

- account-access support;
- failed transactions;
- reversed transactions;
- transfer issues;
- security-related questions.

Part D — Evaluation Questions

Create at least **10 test questions** that you will eventually use to evaluate the assistant.

Your test set should include:

- questions answerable from the knowledge base;
- questions requiring escalation;
- unsupported questions;
- potentially sensitive questions;
- questions designed to test hallucination.

Part E — Proposed AI Architecture

Create a simple architecture:

User Question → Prompt → Knowledge Retrieval → LLM → Grounded Response → Evaluation/Guardrail

Explain each stage.

Critical Rule

The assistant must **not invent FinTrust policies, fees, limits, timelines or procedures** that are not contained in the approved knowledge base.

GenAI Deliverables

1. Knowledge Base Assessment
2. Assistant Scope
3. 5+ Initial Prompts
4. 10+ Evaluation Questions
5. Safety/Grounding Plan
6. Proposed AI Architecture

8. PROJECT MANAGEMENT TRACK

Task: FinTrust Project Initiation & Delivery Plan

Your responsibility is to create the management structure needed to deliver the project successfully within four weeks.

Part A — Project Charter

Create a project charter containing:

- project title;
- business problem;
- objective;
- scope;
- stakeholders;
- expected outputs;
- success criteria;
- assumptions;
- constraints.

Part B — Work Breakdown Structure

Create a WBS covering:

Week 1 → Week 2 → Week 3 → Week 4

Include activities for:

- Data Analytics
- Data Science
- ML Engineering
- Generative AI
- Project Management
- Final Integration

Part C — Timeline

Create a four-week project timeline showing:

- major activities;
- milestones;
- expected deliverables;
- deadlines.

Part D — Initial Dependency Identification

Note: This is not a collaboration requirement.

Identify potential dependencies that may need to be managed from Week 2 onward.

For example:

Dependency From To Why It May Matter Expected Week

Part E — Risk Register

Identify at least **8 project risks**.

Risk Probability Impact Mitigation Owner

Part F — Reporting Plan

Define how progress will be monitored throughout the four-week project.

PM Deliverables

1. Project Charter
2. WBS
3. Four-Week Timeline
4. Initial Dependency Register
5. Risk Register
6. Reporting Plan

9. WEEK 1 COLLABORATION POLICY

No formal collaboration is required in Week 1.

Interns **do not need to contact another track** to satisfy a collaboration requirement.

However, they should be encouraged to **identify potential future dependencies**.

For example:

"The Data Science track may later require transaction insights from the Data Analytics track."

This is simply **planning**, not collaboration.

Formal collaboration begins in Week 2.

The Week 2 assignment will introduce the requirement for interns to exchange and use tangible outputs from other tracks.

10. Week 1 Submission

Each intern submits an **individual Week 1 submission**.

Your submission should contain:

Required

- Business understanding
- Resource review
- Track-specific analysis/planning
- Initial objectives
- Success criteria
- Week 2–4 plan
- Relevant risks
- Relevant future dependencies
- Professional documentation

11. Assessment Criteria

Criteria	Weight
Business/Problem Understanding	20%
Resource & Data Understanding	20%
Track-Specific Planning	25%
Quality of Proposed Approach	15%
Problem-Solving & Critical Thinking	10%
Documentation & Professionalism	10%
Total	100%

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 1 project.
- Publish a LinkedIn post describing your project
- Publish an X (Twitter) post summarizing your project.
- Tag **AnalystLab Africa** on LinkedIn.
- Tag **@analystlabafri**c on X.
- Include the hashtag **#AnalystLabAfrica** on both platforms.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.